CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Code 2: Loops make it move

Goal: I can use a loop to repeat pictures and make an animation.

35 minutes | 7 steps | Computer Science and Programming | Playbook B meeting 27

Ask | Plan | Build | Test | Improve

You are done when

Tell a partner what repeats in your code.

You will use

- Cart laptops, charged
- MakeCode for micro:bit in the browser, simulator only, if approved
- Grid cards for the no-device path

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Open MakeCode — 4 min

- Get your laptop.
- Open the MakeCode link.

2 Step 2: Recap — 1 min

- Tell a partner why order matters.

3 Step 3: Watch a loop — 4 min

- Watch forever repeat the same steps.

4 Step 4: Make it move — 14 min

- Make three pictures that change a little.
- Put them in a forever loop.
- Add a pause between them.

5 Step 5: Debug — 4 min

- Find one thing that looks wrong.
- Fix it and run again.
- Show your animation to a partner.

6 Step 6: Save and shut — 5 min

- Name your project with your initials.
- Close the lid with two hands.
- Put the laptop back and plug it in.

7 Step 7: Tell what repeats — 3 min

- Tell a partner what repeats in your code.

What to turn in

An animation that repeats in a loop.

No computer? Another way to do it

Flip-book grid cards run by a partner three times.

Finished early? Next challenge

Change the pause to speed up or slow down the animation.

Stay safe

Use only the MakeCode link the teacher gives.

Grade targets

Grade 5: Three pictures in a forever loop.
Grade 6: Use a repeat loop with a count and explain how it differs from forever.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.